

ID: 291-0666

NAME: Nowman Siddiqui

QUIZ 7-A

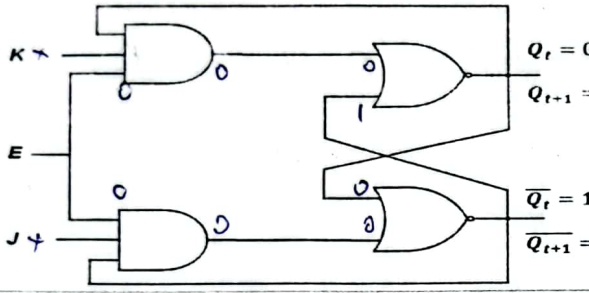
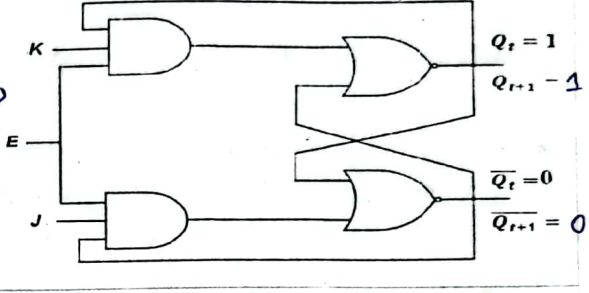
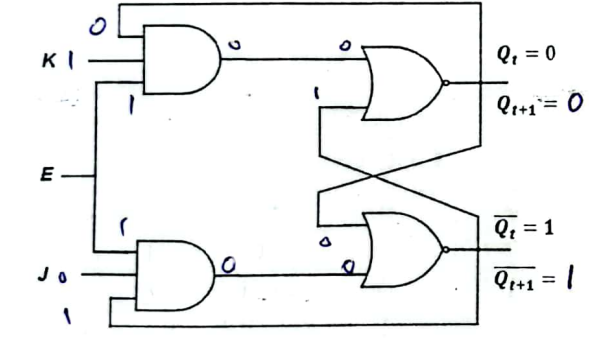
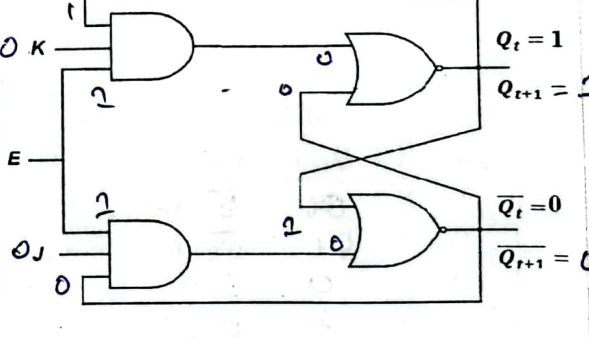
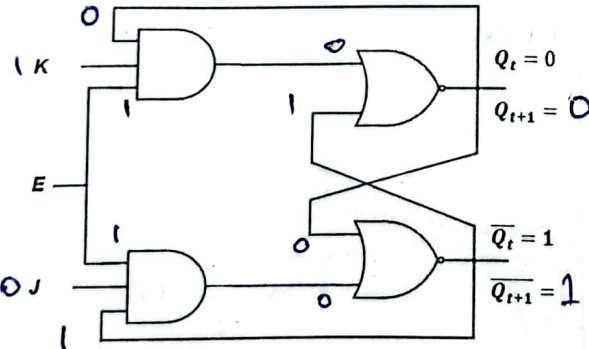
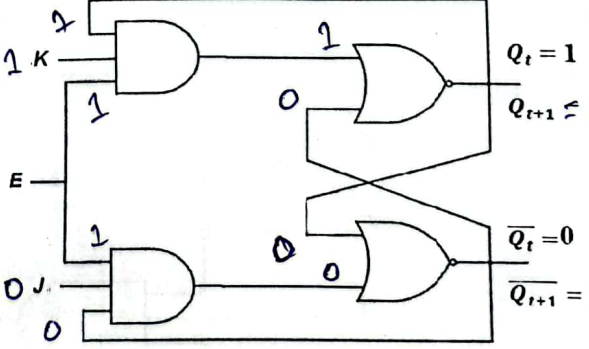
20
20SECTION: E

Instruction: No marks will be awarded for Direct Answers

1. Process the JK Latch given below for all possible cases and fill the table accordingly?

00 1
01 0
10 0
11 0

CIRCUIT:

Case-I ($Q_t = 0$)	Case-II ($Q_t = 1$)
$E=0, J=\times, K=\times, Q_t = 0, Q_{t+1} = \underline{0}$ 	$E=0, J=\times, K=\times, Q_t = 1, Q_{t+1} = \underline{1}$ 
$E=1, J=0, K=0, Q_t = 0, Q_{t+1} = \underline{0}$ 	$E=1, J=0, K=0, Q_t = 1, Q_{t+1} = \underline{1}$ 
$E=1, J=0, K=1, Q_t = 0, Q_{t+1} = \underline{0}$ 	$E=1, J=0, K=1, Q_t = 1, Q_{t+1} = \underline{0}$ 

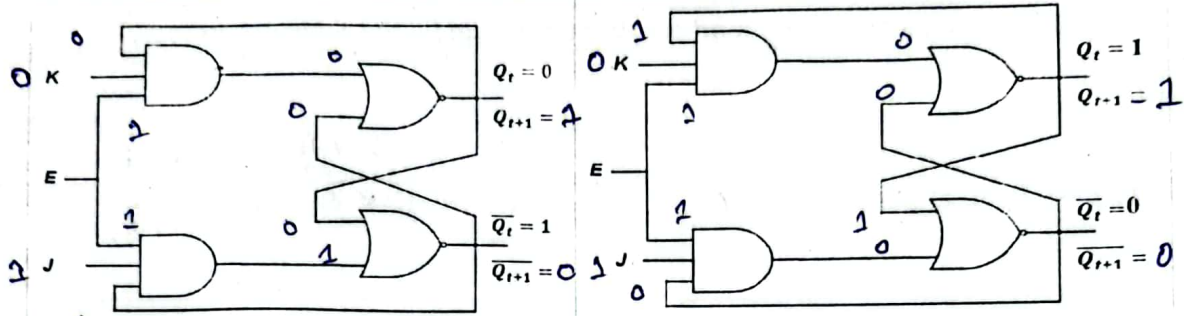
QUIZ 7-A

ID: _____

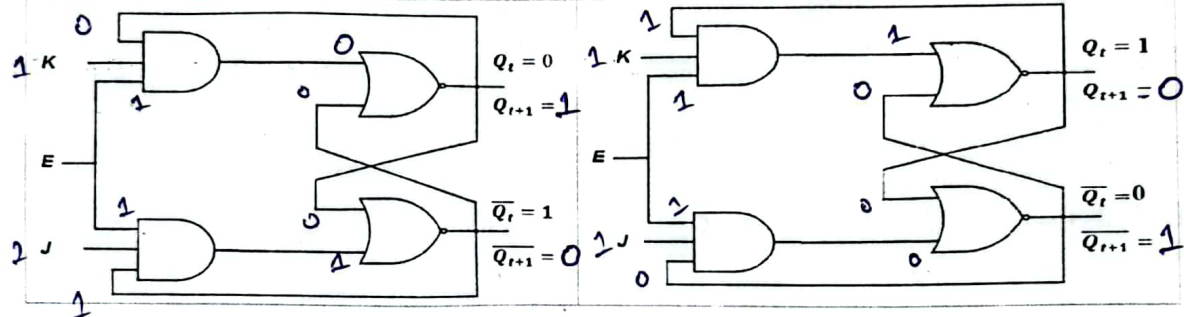
NAME: _____

SECTION: _____

$E=1, J=1, K=0, Q_t=0, Q_{t+1}=\underline{1}$ $E=1, J=1, K=0, Q_t=1, Q_{t+1}=\underline{1}$



$E=1, J=1, K=1, Q_t=0, Q_{t+1}=\underline{1}$ $E=1, J=1, K=1, Q_t=1, Q_{t+1}=\underline{0}$



JK Latch					JK LATCH with SET/RESET				
E	J	K	Q_{t+1}	$\overline{Q}_{t+1}$	SET	CLEAR	Q_{t+1}	$\overline{Q}_{t+1}$	
0	X	X	Q_t	$\overline{Q}_t$	0	0	Q_{t+1}	$\overline{Q}_{t+1}$	✓
1	0	0	Q_t	$\overline{Q}_t$	0	1	0	1	✓
1	0	1	0	1	1	0	1	0	✓
1	1	0	1	0	1	1	Invalid	Invalid	✓
1	1	1	$\overline{Q}_t$	Q_t					

2. Modify given NOR Based JK Flip Flop circuit to introduce Asynchronous CLEAR and SET inputs fill table above?

